## Progressive exercises

**E1 · Direct application.** Calculate $\displaystyle\int_0^2 (3x^2 - 2x + 1)\,\mathrm{d}x$ by finding an antiderivative.

**E2 · Application and justification.** In an orthonormal coordinate system, calculate the area of the region bounded by the curves $y = x$ and $y = x^2$, between x = 0 and x = 1. Give the result in units of area.

**E3 · Method to organise.** Calculate the mean value of the exponential function over the interval $[0;2]$.

**E4 · Choice of strategy.** Using integration by parts, calculate $\displaystyle\int_1^e \ln x\,\mathrm{d}x$.

**E5 · Synthesis or proof.** For every $n \in \mathbb{N}$, set $I_n = \displaystyle\int_0^1 \frac{x^n}{1+x}\,\mathrm{d}x$, with $x^0 = 1$ in the integrand. Prove that $0 \leq I_n \leq \frac{1}{n+1}$, then deduce the limit of $(I_n)$.